

Write legibly showing all the steps VERY clearly. (Total points = 100).

- (1) Let $(\Omega, \mathcal{F}, P) = ((0, 1), \mathcal{B}((0, 1)), m)$, where m denotes the Lebesgue measure. Define the random variables $X, \{X_n\}$: For each $\omega \in \Omega$, $X(\omega) = 1$ and

$$X_1(\omega) = 0, X_n(\omega) = 1 + (n-1)\mathbf{I}_{\{(0, \frac{1}{n}]\}}(\omega), n \geq 2.$$

- (a) Does $X_n \rightarrow X$ a.s. (P) as $n \rightarrow \infty$?
- (b) Does $X_n \rightarrow X$ in probability as $n \rightarrow \infty$?
- (c) Does $X_n \rightarrow X$ in L^p as $n \rightarrow \infty$? Argue for different values of $p \in (0, \infty)$.
- (d) Is $\{X_n\}$ a uniformly integrable sequence of random variables?

(10+10+10+10=40 points)

- (2) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $g, f, f_n \in L^1(\Omega, \mathcal{F}, \mu) \equiv L^1$ for $n \geq 1$.

- (a) If $f_n \leq f$ then show that $\limsup_{n \rightarrow \infty} \int f_n d\mu \leq \int (\limsup_{n \rightarrow \infty} f_n) d\mu$.
- (b) Assuming $f > 0$ a.e. (μ) and $\frac{1}{f} \in L^1$, show $\left(\int f d\mu\right) \left(\int \frac{1}{f} d\mu\right) \geq 1$.
- (c) If $\int_A f d\mu = \int_A g d\mu$ for all $A \in \mathcal{F}$, then show that $f = g$ a.e. (μ).

(10+10+10=30 points)

- (3) Let $(\Omega, \mathcal{F})$ be a measurable space and μ, μ_1, μ_2, ν are some σ -finite measures on it.

- (a) If $\mu \ll \nu$ and $\mu \perp \nu$, show: μ is the zero measure, i.e. $\mu(A) = 0$, for all $A \in \mathcal{F}$.

Let $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\mu_3 \equiv \mu_1 + 3\mu_2$, where $\mu_1 = \text{Normal}(-1, 1)$, $\mu_2 = \text{Bernoulli}(\frac{1}{5})$ and $\nu = m$ is the Lebesgue measure.

- (b) Find the Lebesgue decomposition of μ_3 w.r.t. ν , and find the Radon Nikodym derivative $\frac{d\mu_{3a}}{d\nu}$, where μ_{3a} is the absolutely continuous component of μ_3 in the decomposition.

- (c) If $g(x) = x$, for $x \in \mathbb{R}$, and μ_{3a} is as in (d), find $\int g d\mu_{3a}$.

Give the final answer in numerical form. You can state and use any calculus/basic statistics facts without proofs (but explain the *measure-theoretic* steps.).

(10+10+10=30 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0, 1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi\sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi}\sigma} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$